

MONITORING CONICAL INTERSECTION DYNAMICS WITH QUANTUM DYNAMICS AND X-RAY SPECTROSCOPY

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- Spectroscopy
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Linda Young

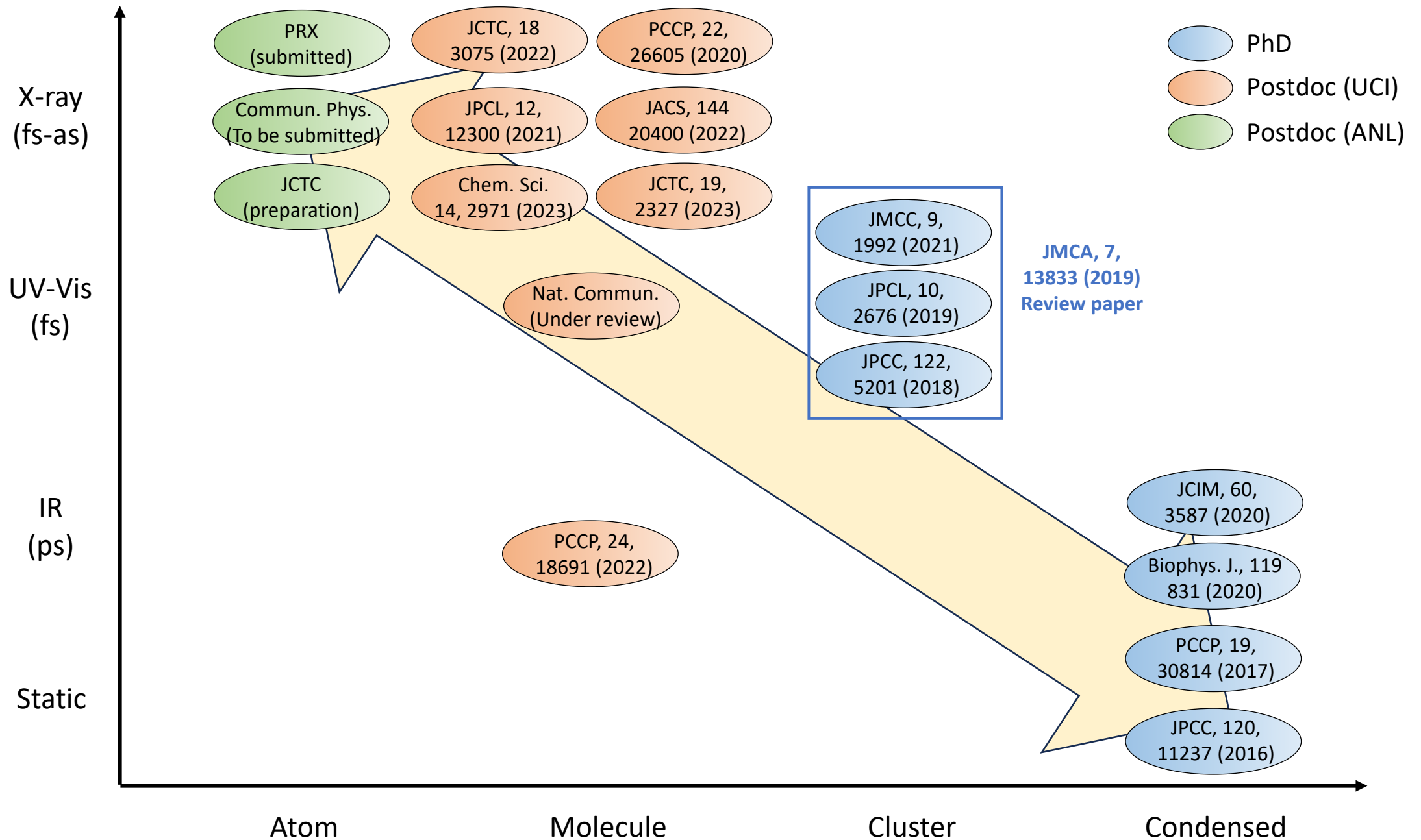
- Exp + Theory
- AMO Physics

BS: Chem. E
Ph.D.



Jin Yong Lee

- Photocatalysis
- DFT



Research Topics

Molecular Magnet

- Diradical organic magnet
- MoS2 inorganic magnet
- Electric field control

TiO2 Photocatalysis

- Charge carrier dynamics
- Size and Shape (facet)
- Oxygen defect & Doping

Histidine Tautomerism

- A β protein misfolding
- Tau protein misfolding
- 2DIR probe / Classical MD

Molecular Chirality

- Global vs local chirality
- E1-E2 vs E1-M1 at X-ray
- Pulse shaping, OAM beam

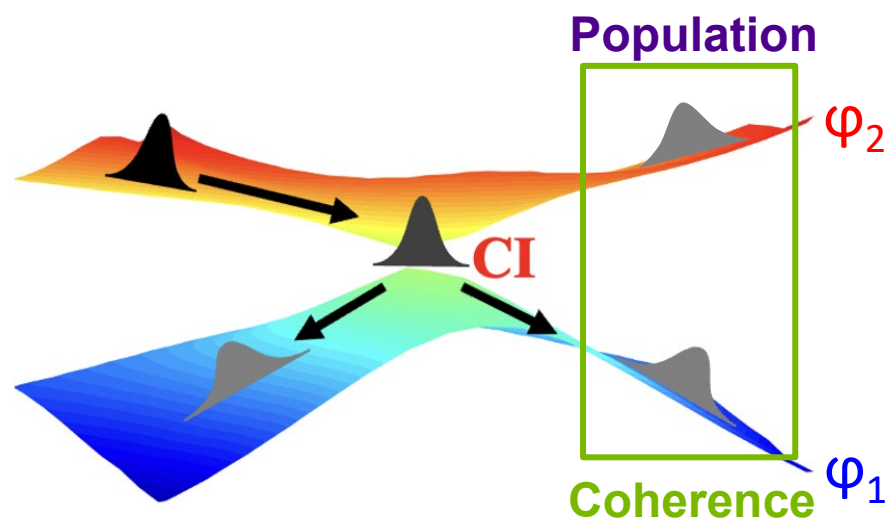
Conical Intersections

- Quantum nuclear dynamics
- X-ray absorption / Raman / Diffraction
- Photoelectron spectroscopy

X-ray Radiation Damage

- Multi-ionization of ICI
- CF4 Coulomb explosion
- $\text{CO}^{*+} \rightarrow \text{CO}^{2+}$ vs $\text{C}^+ + \text{O}^+$

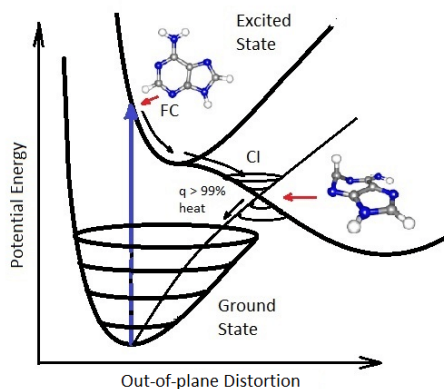
Conical Intersection (CI)



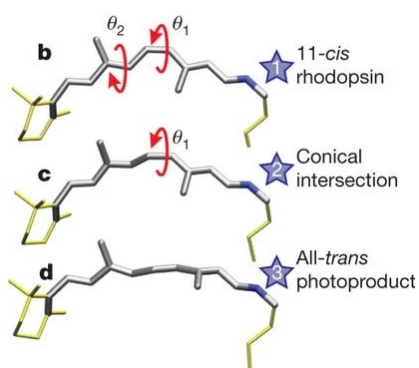
- Potential energy surfaces cross
- **Non-radiative** population transfer
 - At tens to hundreds of femtoseconds
 - Outcompetes radiative relaxation channels

$$\psi = c_1\phi_1 + c_2\phi_2 \quad \leftarrow \begin{array}{l} \text{---} \phi_2 \\ \text{---} \phi_1 \end{array}$$

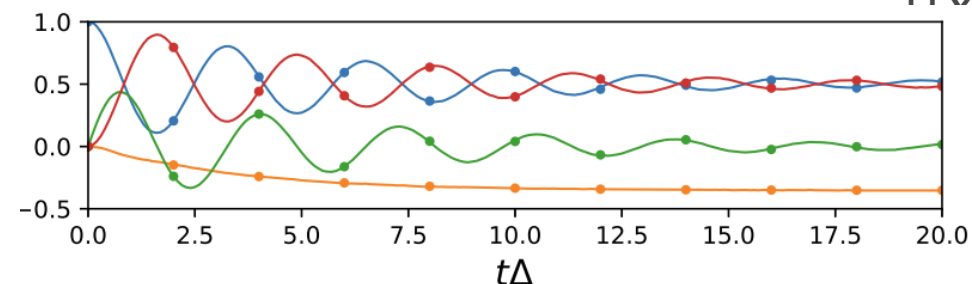
Protection against DNA damage



Photoisomerization in vision



- **Population** ($c_1c_1^*\phi_1\phi_1^*$ & $c_2c_2^*\phi_2\phi_2^*$) $\rightarrow$ ATAS
- **Coherence** ($c_1c_2^*\phi_1\phi_2^* + c_2c_1^*\phi_2\phi_1^*$) $\rightarrow$ TRUECARS TRXD



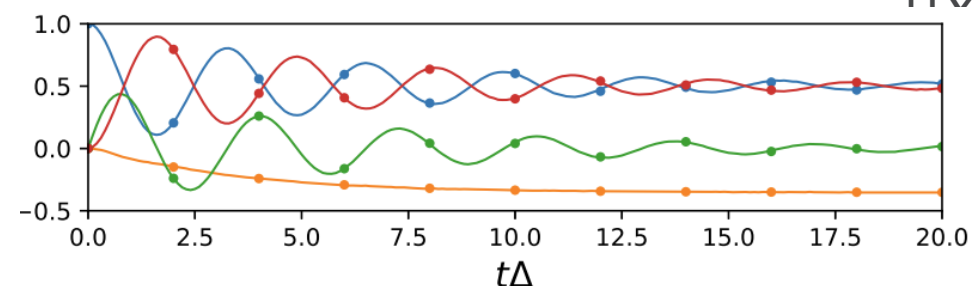
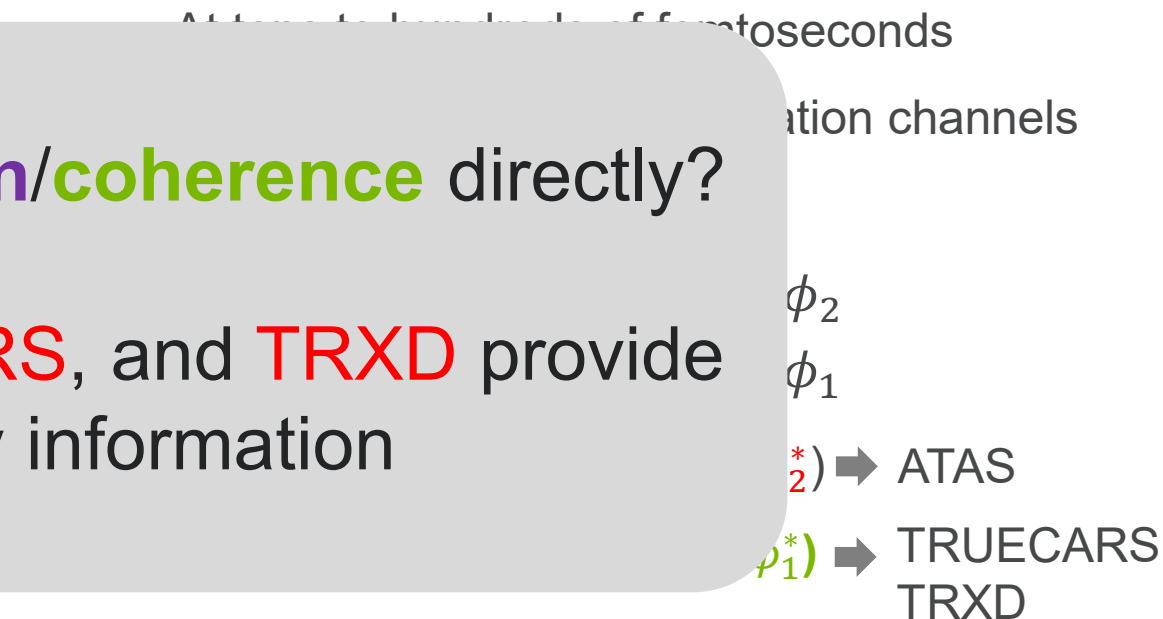
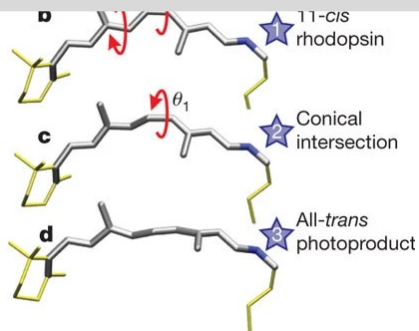
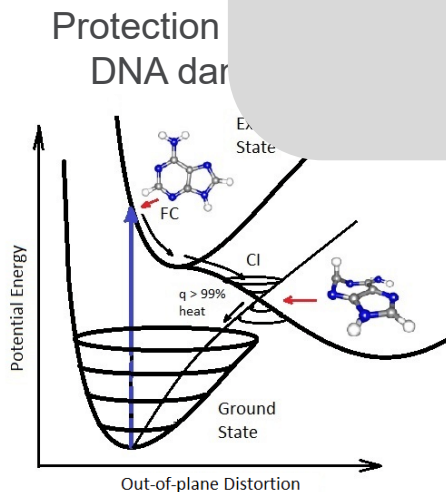
Ullah A, et al., doi:10.26434/chemrxiv-2023-0xkv1-v2

Conical Intersection (CI)



- Potential energy surfaces cross
- **Non-radiative** population transfer

1. Can we probe **population/coherence** directly?
2. How do **ATAS**, **TRUECARS**, and **TRXD** provide complementary information



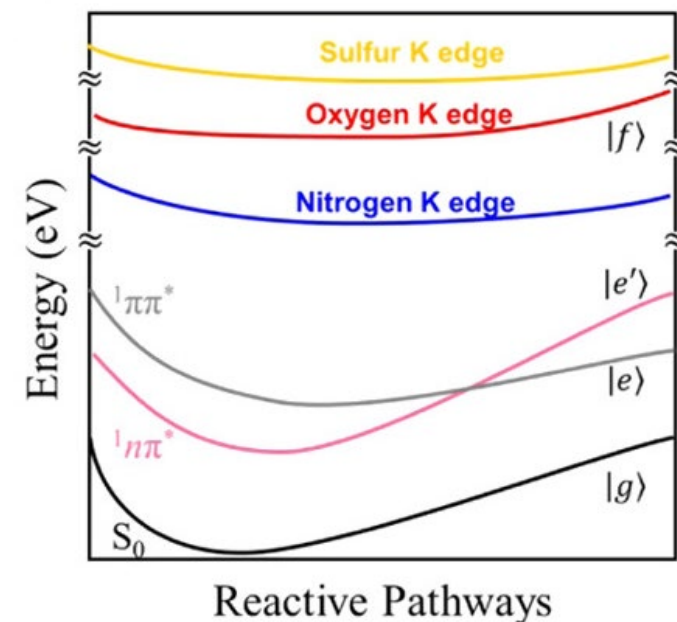
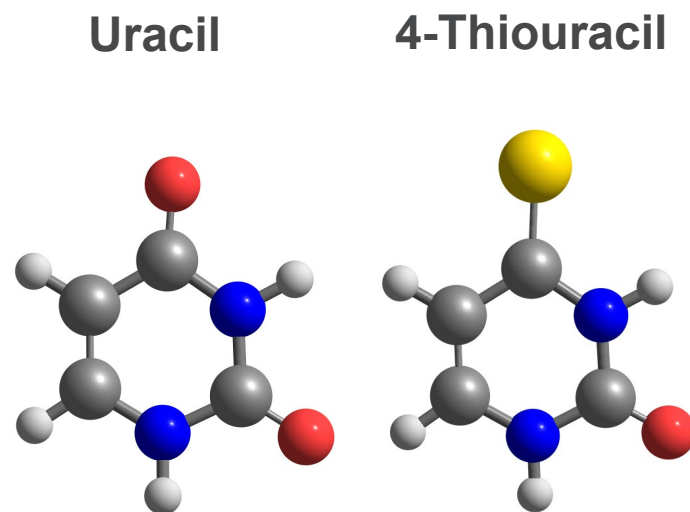
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$S_2 \rightarrow S_1$ Conical intersection in gas-phase 4-thiouracil

Visualizing conical intersection passages via vibronic coherence maps generated by stimulated ultrafast X-ray Raman signals

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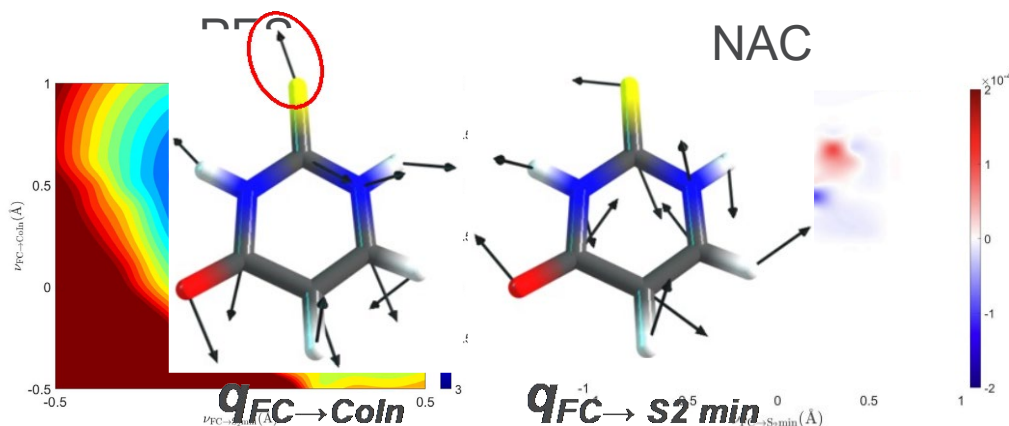


- Uracil, a nucleobase in RNA transcription
- No significant ISC (0.016 at 230 nm)
- Ultrafast relaxation through conical intersection
 - $S_2 \rightarrow S_1$ at ~ 200 fs
- 4-Thiouracil, an O is substituted by S from Uracil
- High triplet quantum yield: $S_1 (n\pi^*) \rightarrow T_{1/2}$ Intersystem crossing
- However, $S_2 (\pi\pi^*) \rightarrow S_1 (n\pi^*)$ internal conversion **mechanism is very unknown**

Computational Details

Electronic structure theory

- **SS-CASPT2/CASSCF(20e/16o)**
 - 1s core orbital in RAS1 (max 1 h)
 - 9 occupied orbitals
 - 6 unoccupied orbitals (max 4 e-)
- 3 states ($S_0 - S_2$) for valence
- 30 states for S / 10 states for N & O
- PES (V_{ii}), V_{NAC} , TDM (μ), Kinetic energy (T) are discretized on 128 x 128 points.



Grid-based Quantum Dynamics

- 2D reactive coordinate

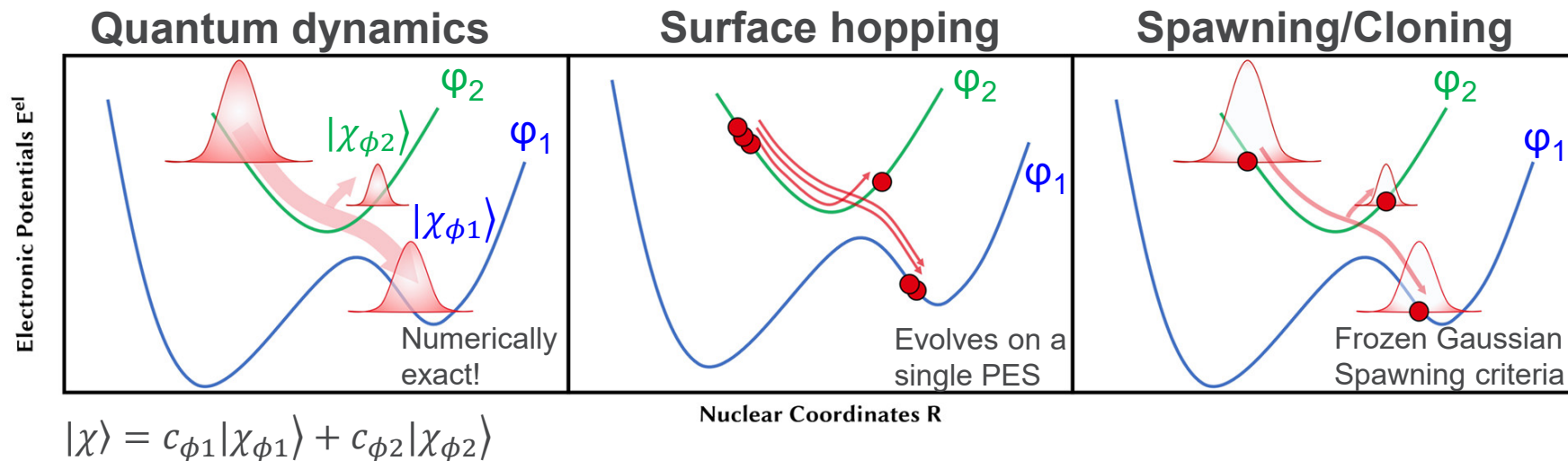
$$i\hbar \frac{\partial}{\partial t} \begin{bmatrix} \chi_1(t) \\ \vdots \\ \chi_n(t) \end{bmatrix} = \begin{bmatrix} H_{11} & \dots & H_{1n} \\ \vdots & \ddots & \vdots \\ H_{n1} & \dots & H_{nn} \end{bmatrix} \begin{bmatrix} \chi_1(t) \\ \vdots \\ \chi_n(t) \end{bmatrix}$$

$$H_{ii} = T + V_{ii}(R)$$

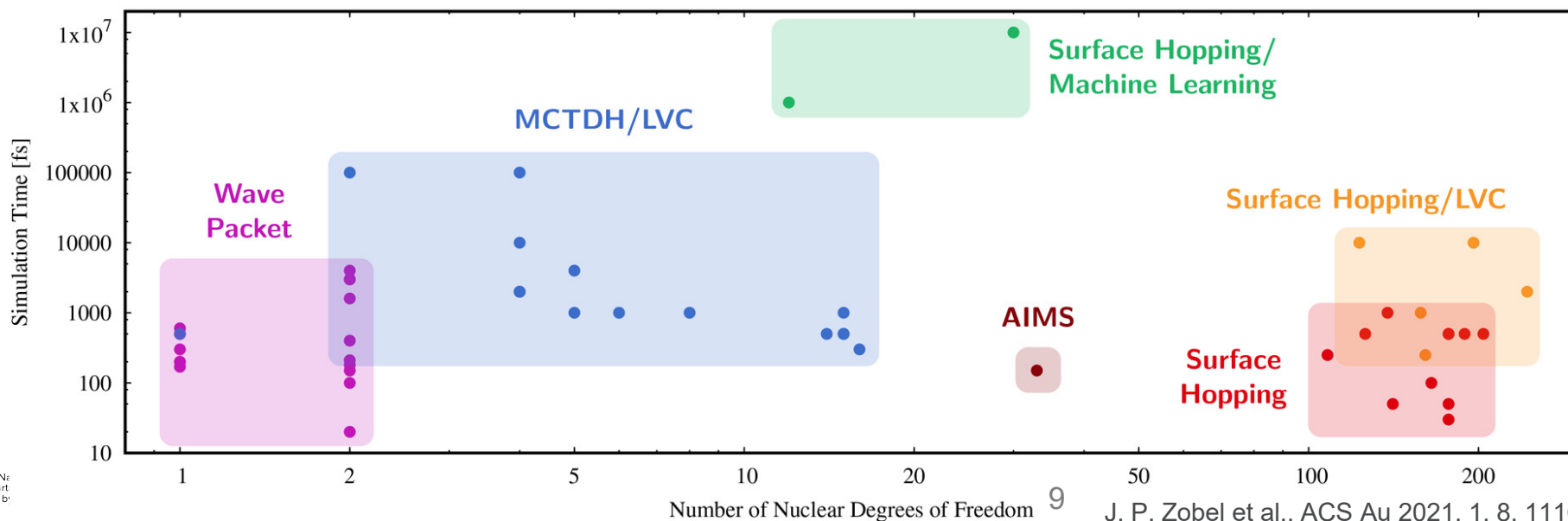
$$H_{ij} = V_{NAC} + V_{SOC} - \mu E_{pu}(t)$$

- WP propagates by solving **the time-dependent Schrodinger equation exactly** (numerically) but on reduced dimension (2D/3D)
- **Quantum effect** of nuclei: decoherence, tunneling are included

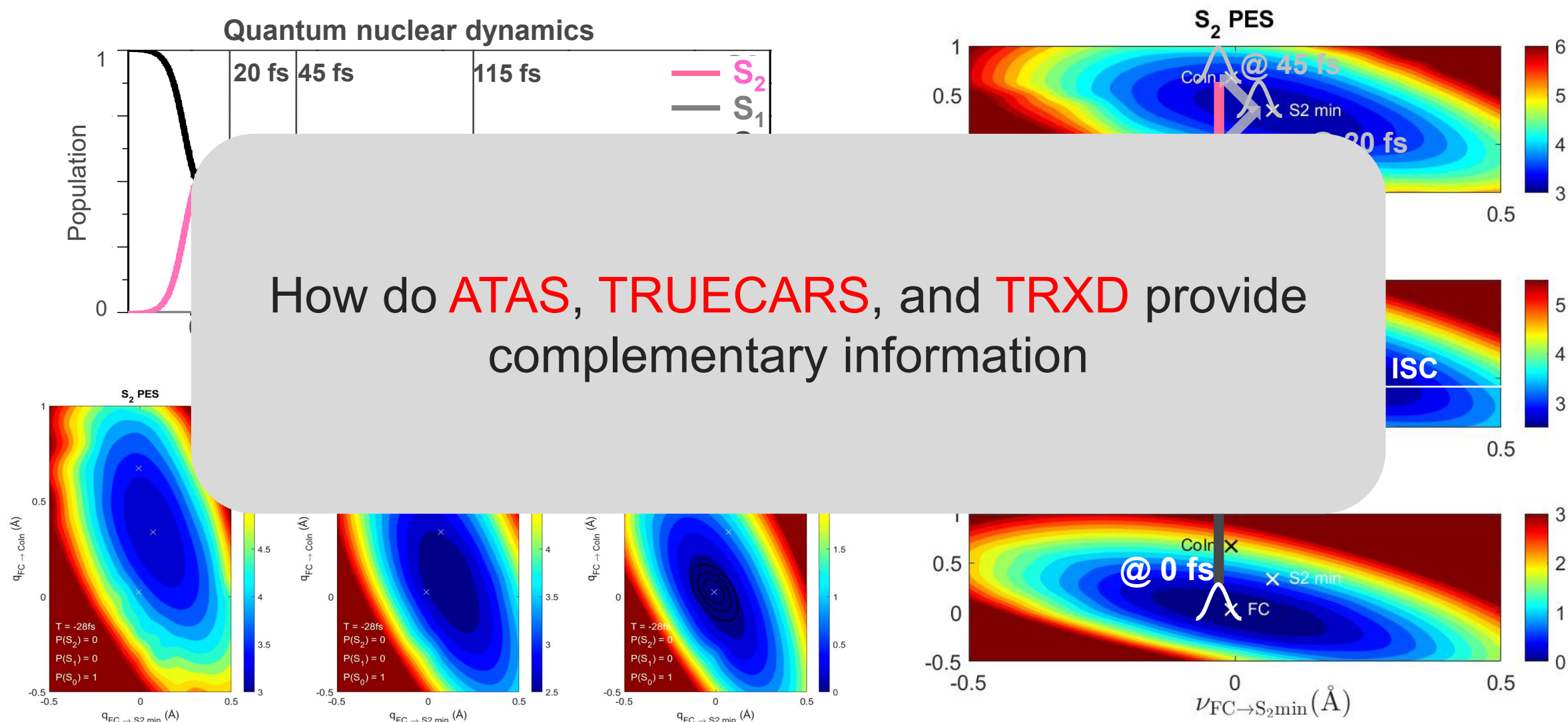
Grid-based Exact Quantum Dynamics



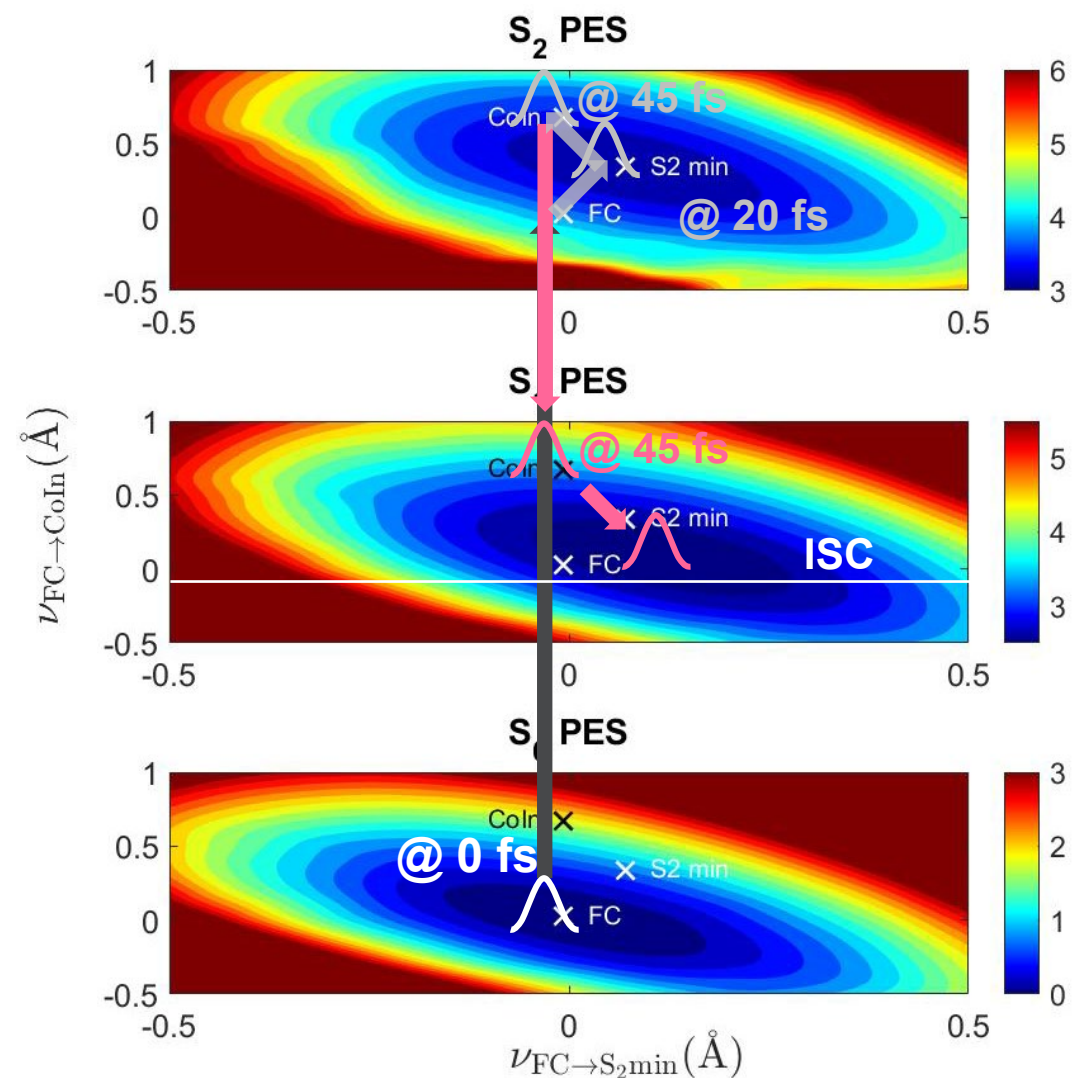
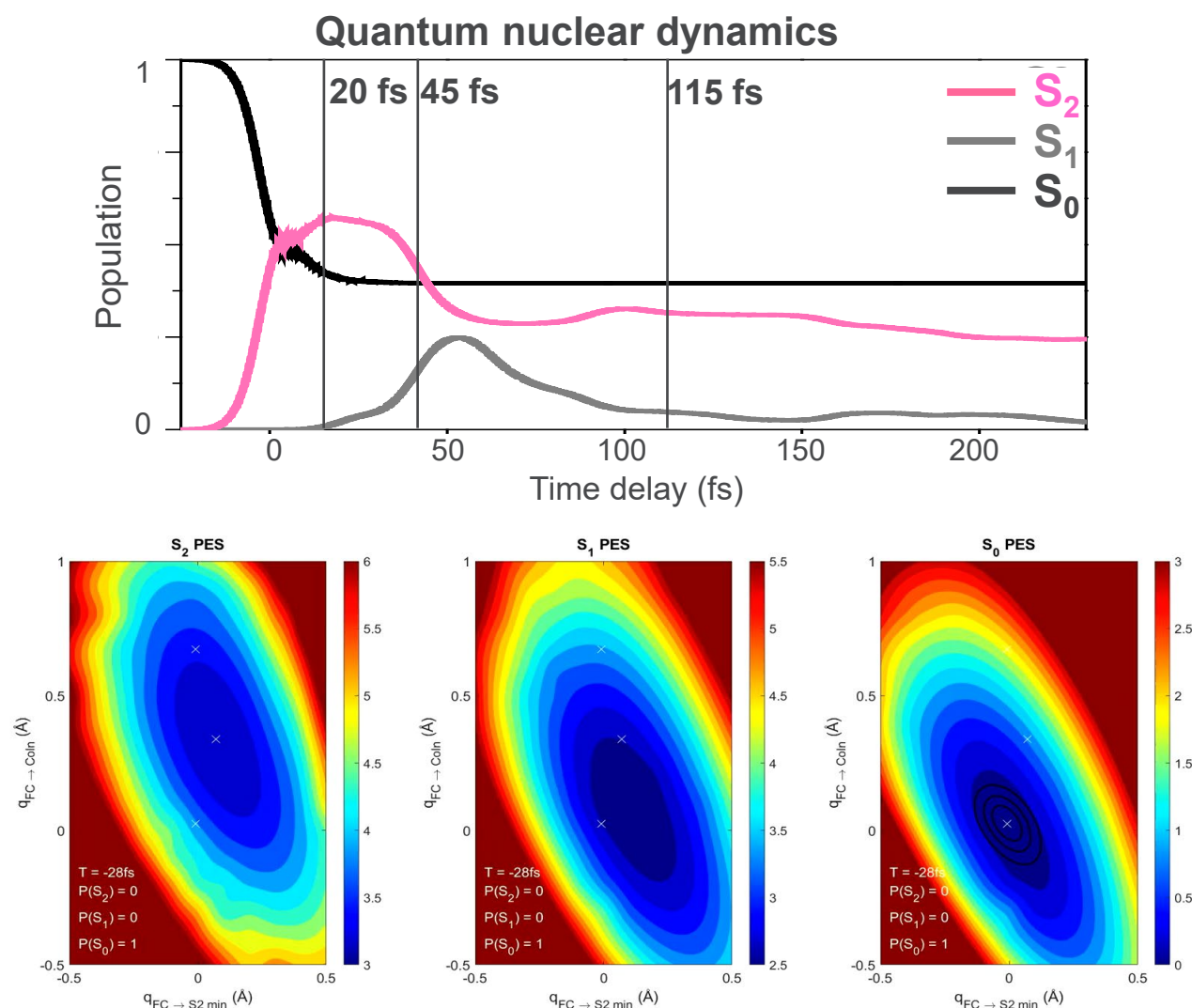
Simulation Times in Nonadiabatic Dynamics Studies



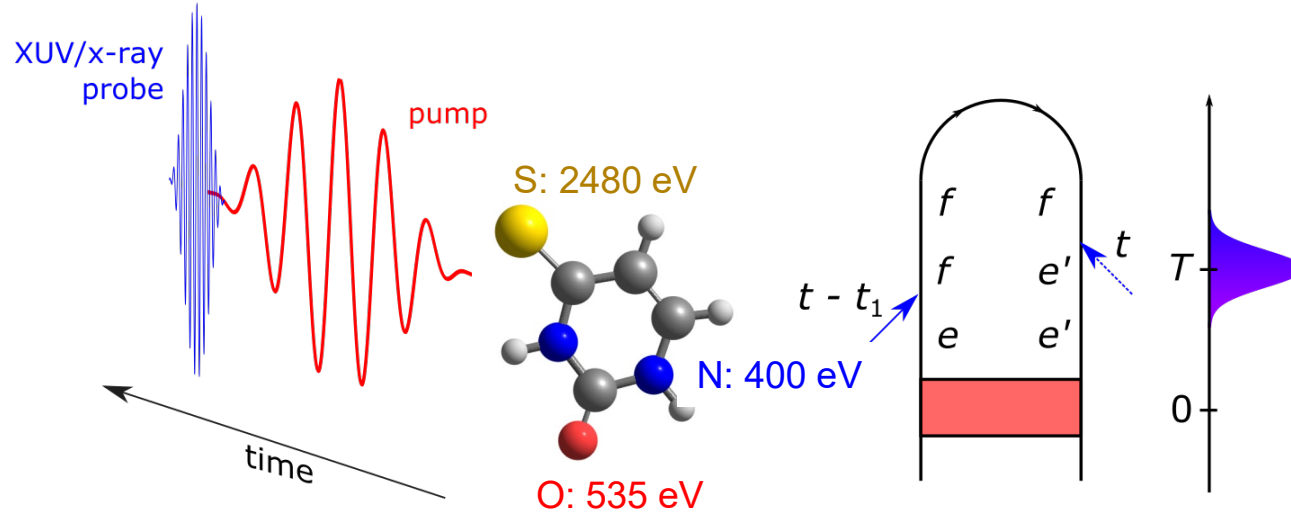
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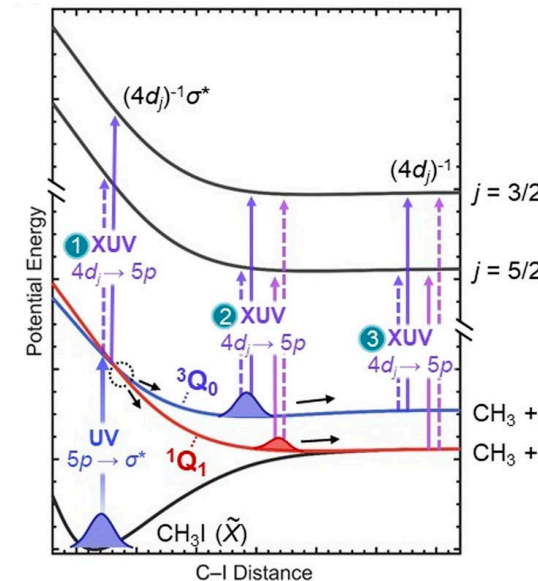


Attosecond transient absorption spectroscopy (ATAS)

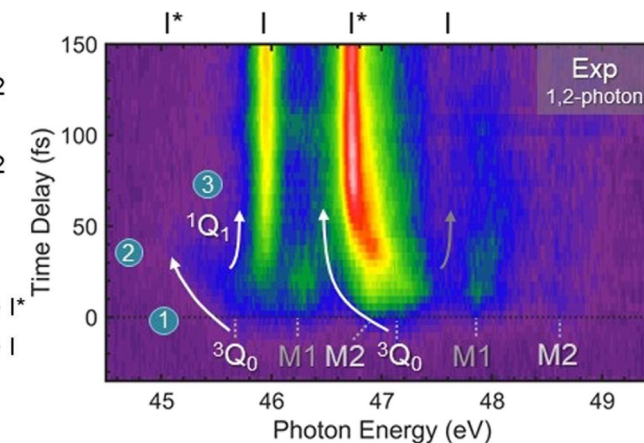


- In **ATAS**, the probe pulse is **resonant** with specific high-energy transitions in the material, due to, e.g., inner-core states.

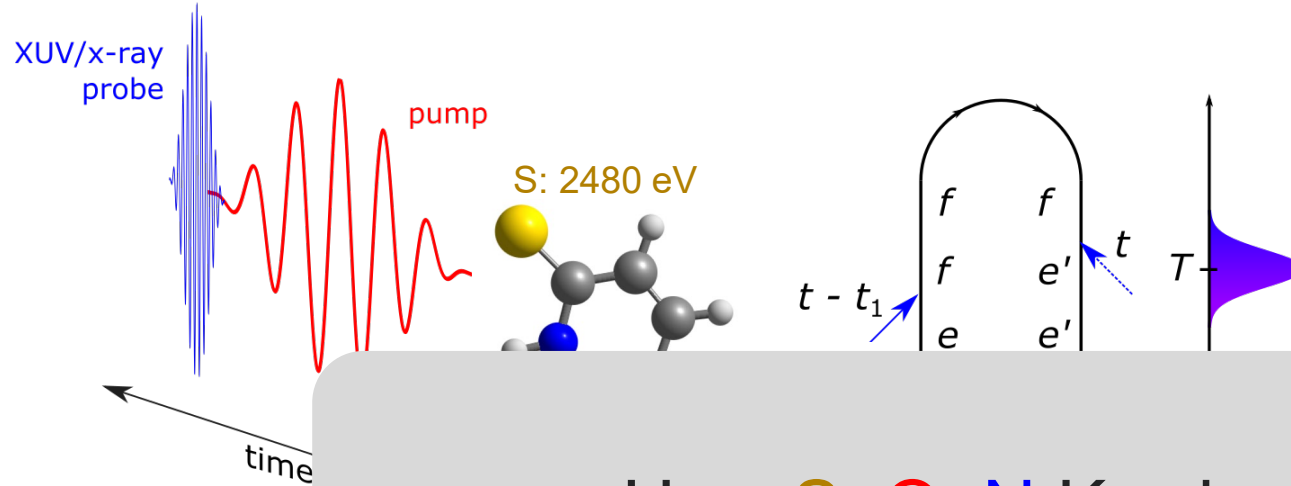
- Attosecond **X-ray** version of transient absorption spectroscopy
- Element/site specific core-to-valence transition** (absorption profile) encodes the transient dynamics of the target system
- S, O, N** atom: site specific photoprobe
- The nonadiabatic passage is inferred from the **appearance /splitting of absorption lines (population)**



ATAS of **CH₃I** passing conical intersection



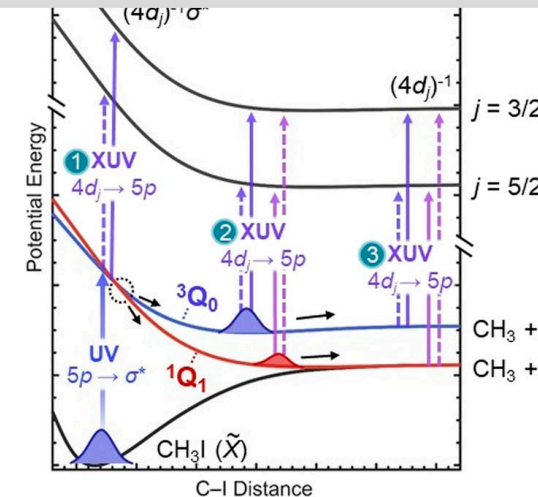
Attosecond transient absorption spectroscopy (ATAS)



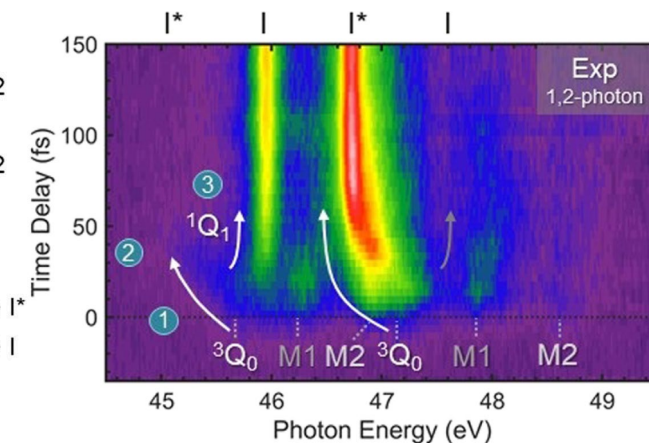
- In **ATAS**, the probe pulse is **resonant** with specific high-energy transitions in the material, due to, e.g., inner-core states.

How **S**, **O**, **N** K-edges are different?

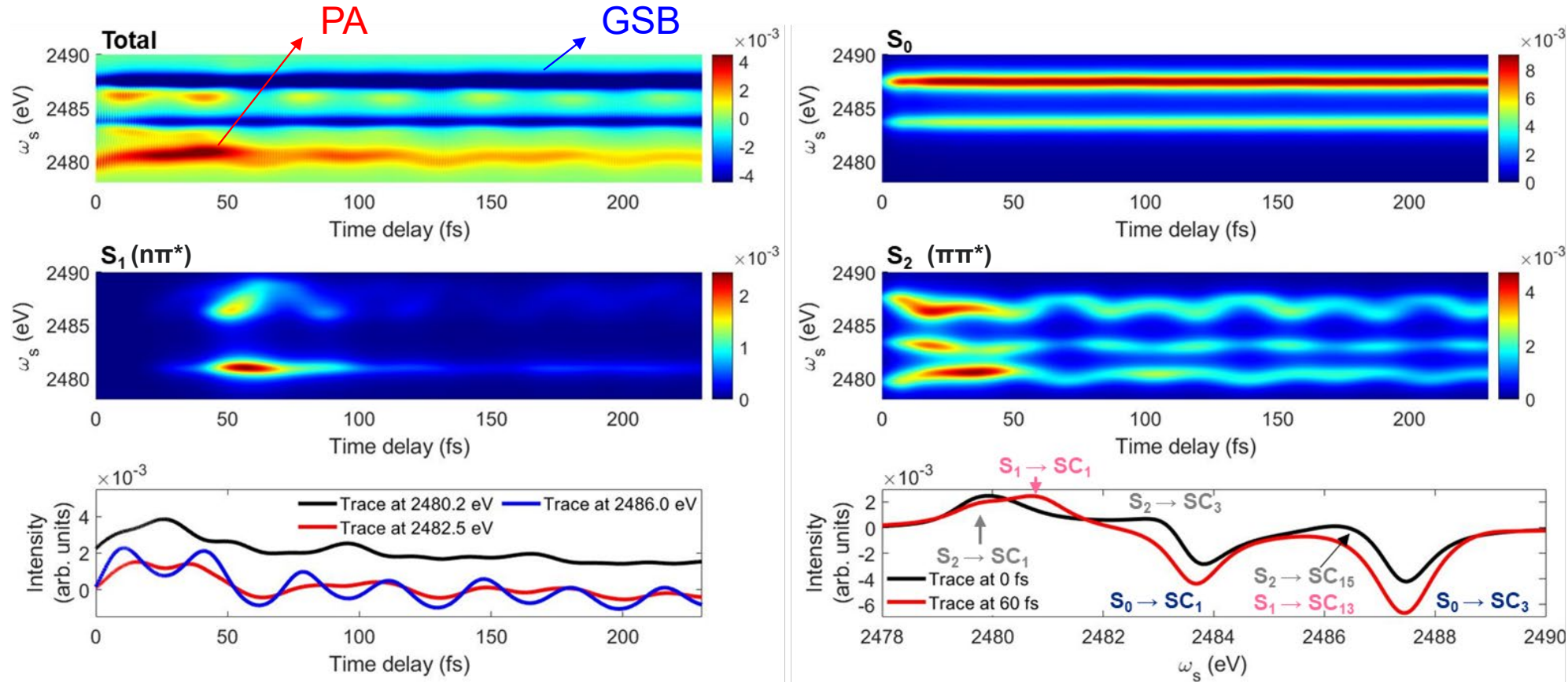
- Attosecond X-ray spectroscopy
- Element/site specific core-to-valence transition** (absorption profile) encodes the transient dynamics of the target system
- S**, **O**, **N** atom: site specific photoprobe
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of CH_3I passing
local intersection

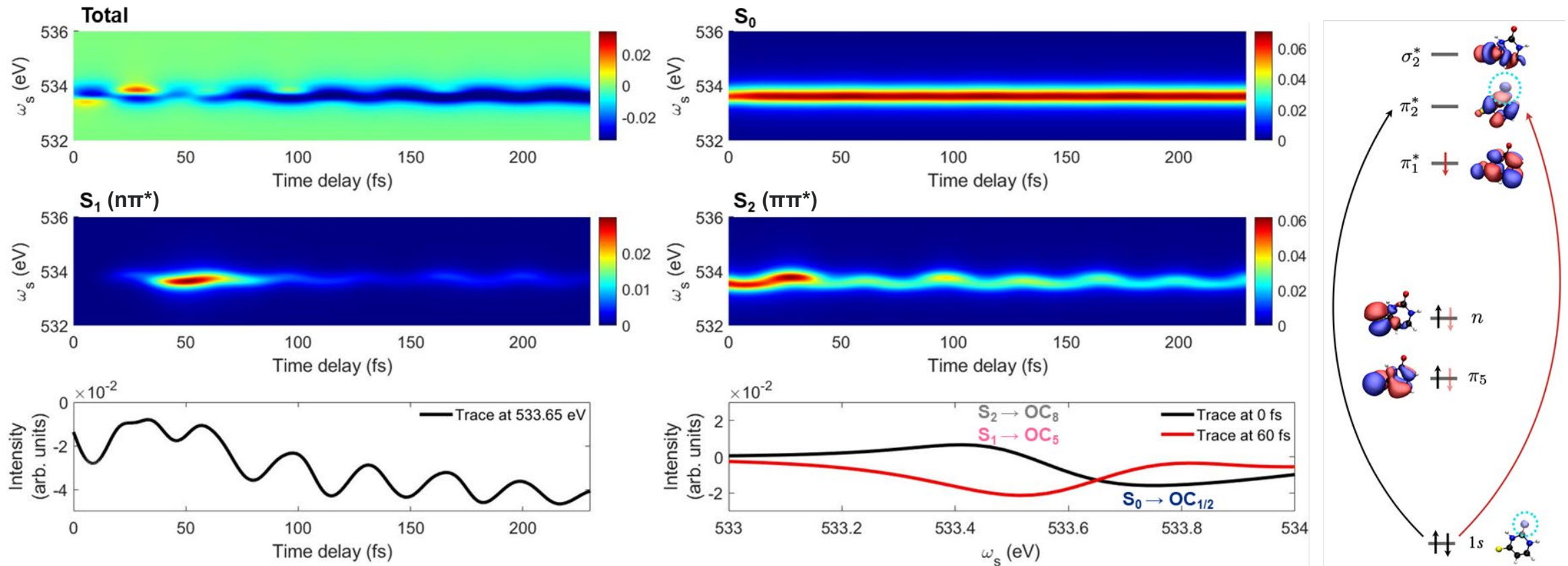


ATAS - S K-edge



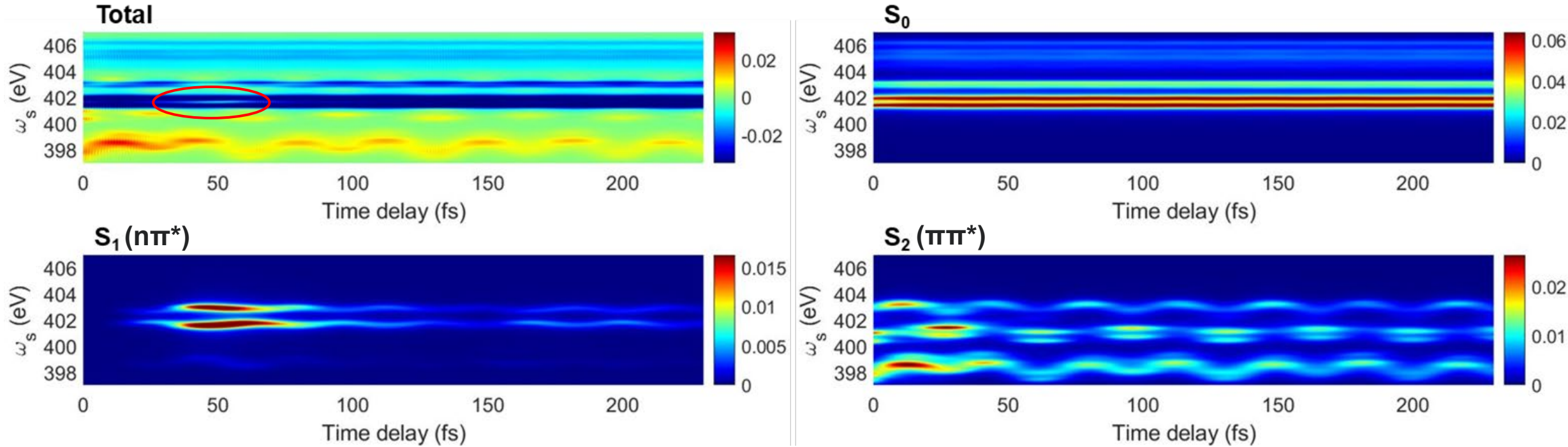
- The $\pi\pi^*$ and $n\pi^*$ have semi-vacant π_5 and n orbitals
- The pre-edge at 2480.2 eV shows background free signal and sensitive from the $\pi\pi^*$ and $n\pi^*$
- The large lifetime broadening of S 1s edge (**0.613 eV**)

ATAS - O K-edge



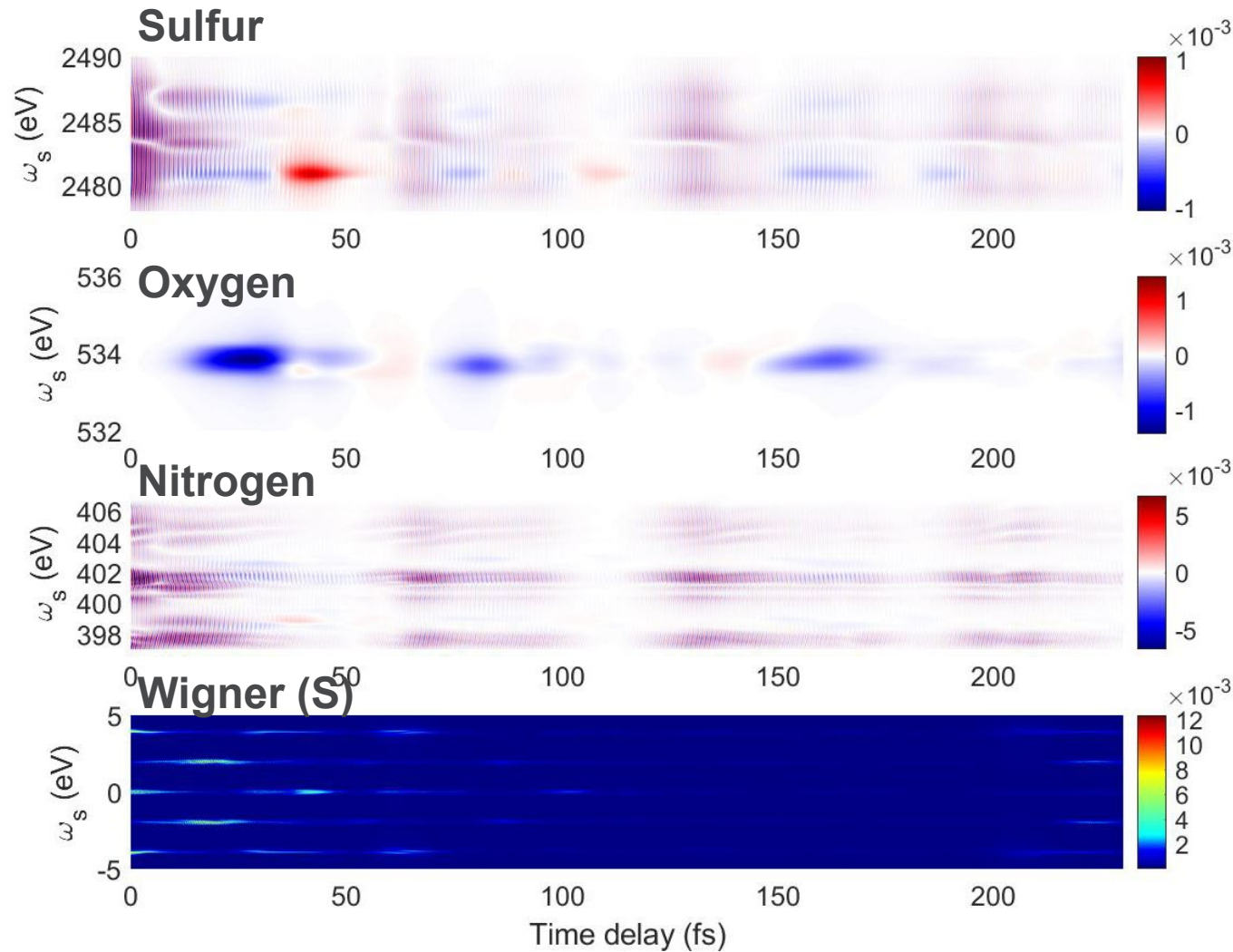
- All core-to-valence excitations are associated with $1s \rightarrow \pi_2^*$ thus insensitive to the molecular deformation
- The $\pi\pi^*$ and $n\pi$ have one bright core state each, OC_5 and OC_8 respectively, but overlap with $S_0 \rightarrow OC_{1/2}$.
- O 1s-edge provides steady transient absorption features without spectral dynamics

ATAS - N K-edges



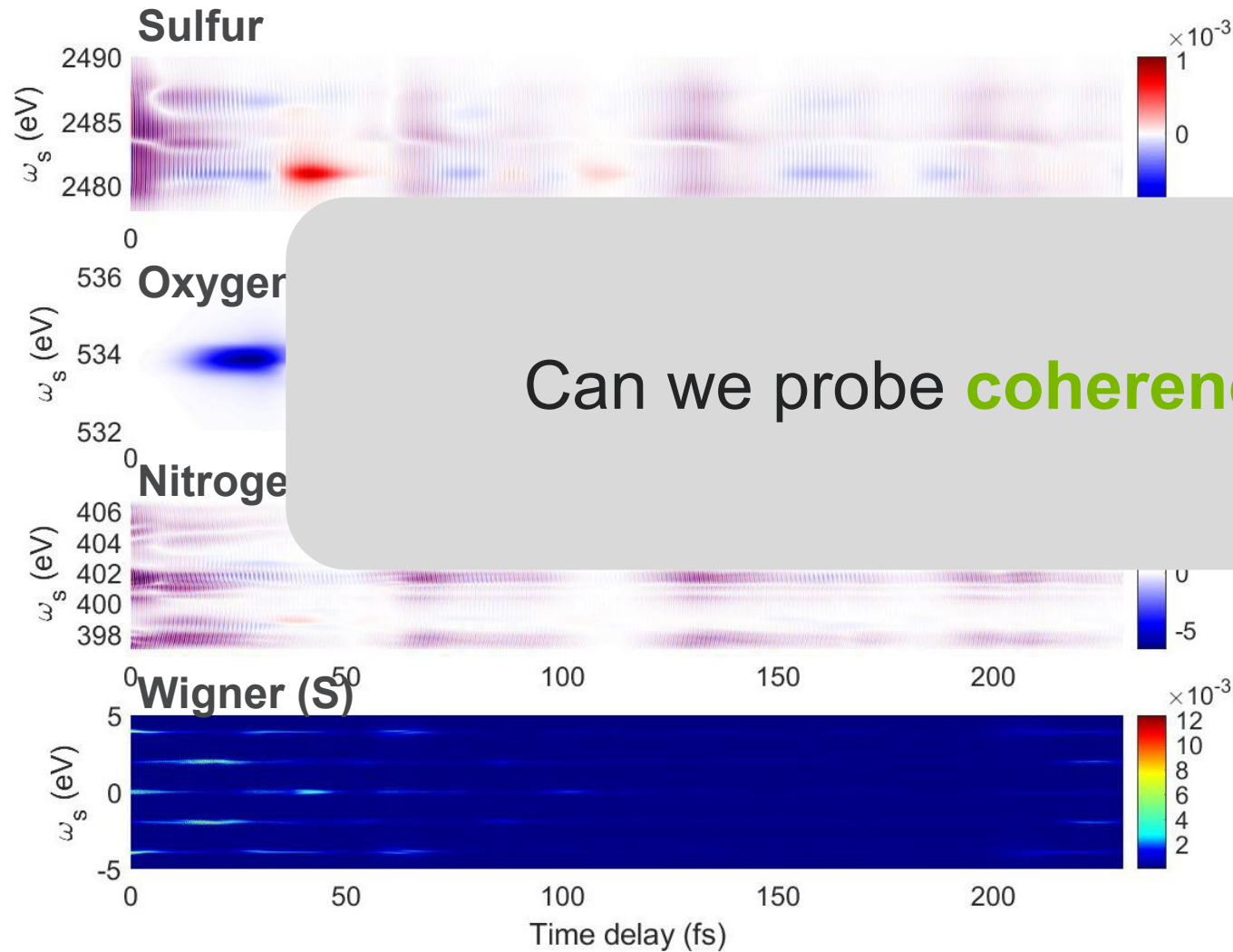
- N has much narrower lifetime broadening – **0.132 eV** (S: 0.613 eV)
- Pre-edge at 398~399 eV is dominated by the ππ* transitions
- nπ* arises at around 401.7 eV
- The depletion/recovery of the GSB indirectly probes the population of the nπ* and its ISC

ATAS - Coherence contribution?



- ATAS involves contribution from electronic coherence ($e \neq e'$)
- Coherence magnitude is weaker than the populations, hence **masked** in the total signal

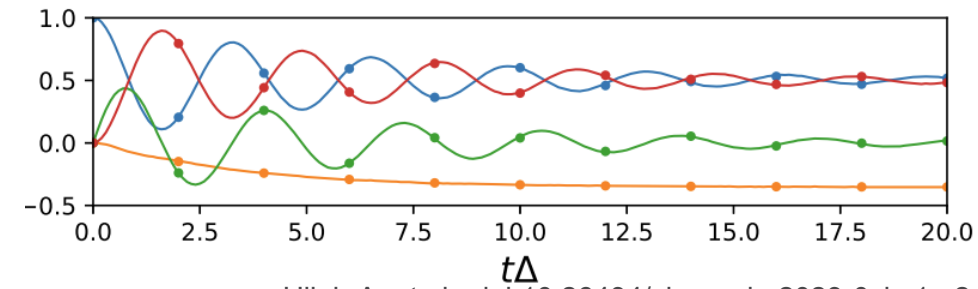
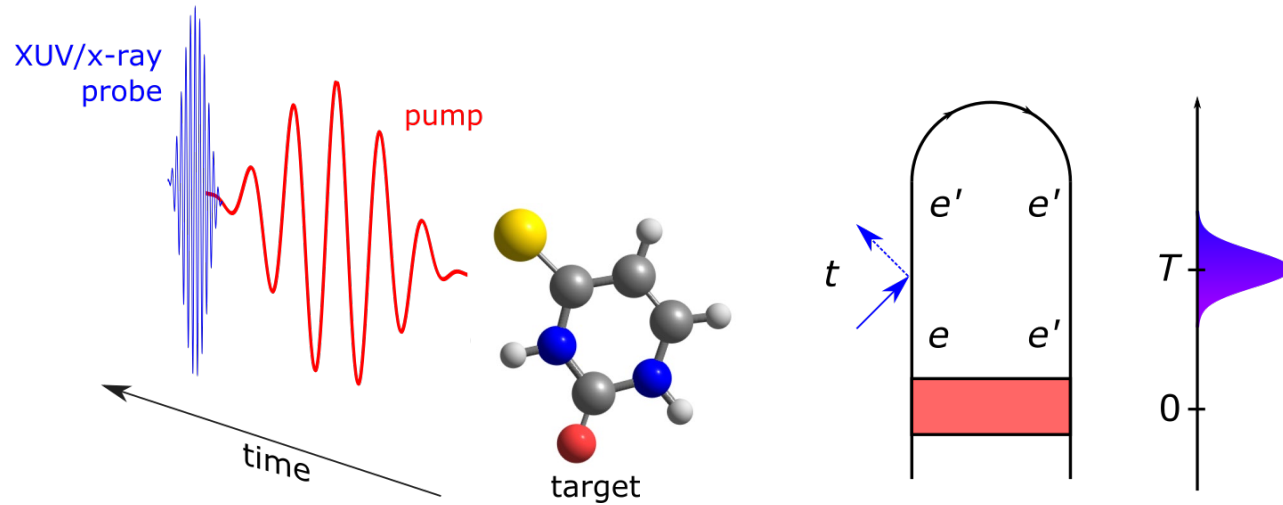
ATAS - Coherence contribution?



- ATAS involves contribution from electronic

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TRUECARS (Transient redistribution of ultrafast electronic coherences in attosecond Raman signals) – X-ray version of hybrid CARS



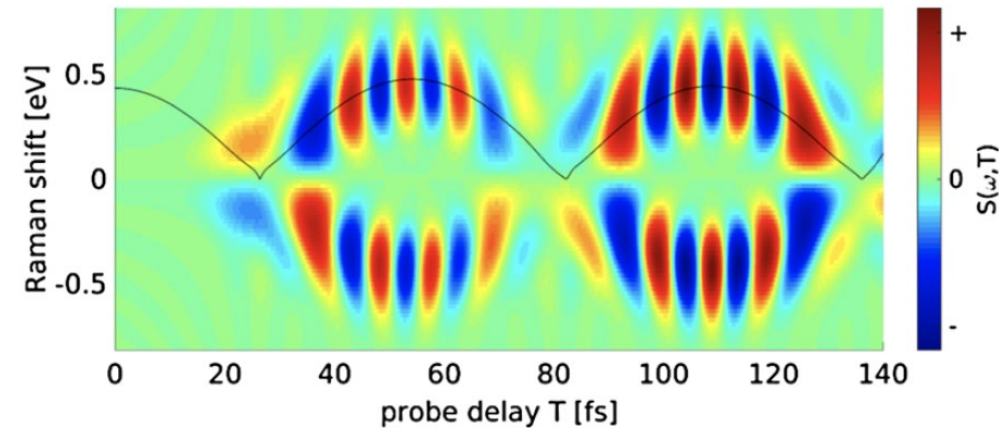
Ullah A, et al., doi:10.26434/chemrxiv-2023-0xkv1-v2

- Pump – **hybrid / off-resonant** X-ray probe
 - Narrowband – temporal resolution
 - Broadband – broad spectral bandwidth
 - Off-resonance → linearity of polarizability

$$\begin{aligned}
 S(\omega_s, T) &= -2\text{Im}\left\{ \tilde{\mathcal{E}}_B^*(\omega_s - \omega_{B0}) \int dt \mathcal{E}_N(t - T) \right. \\
 &\quad \times e^{i(\omega_s - \omega_{N0})(t-T)} \sum_{ee'} \left. \tilde{c}_{e'}^*(t) c_e(t) \langle \chi_{e'}(t) | \hat{\alpha}_{e'e} | \chi_e(t) \rangle \right\}
 \end{aligned}$$

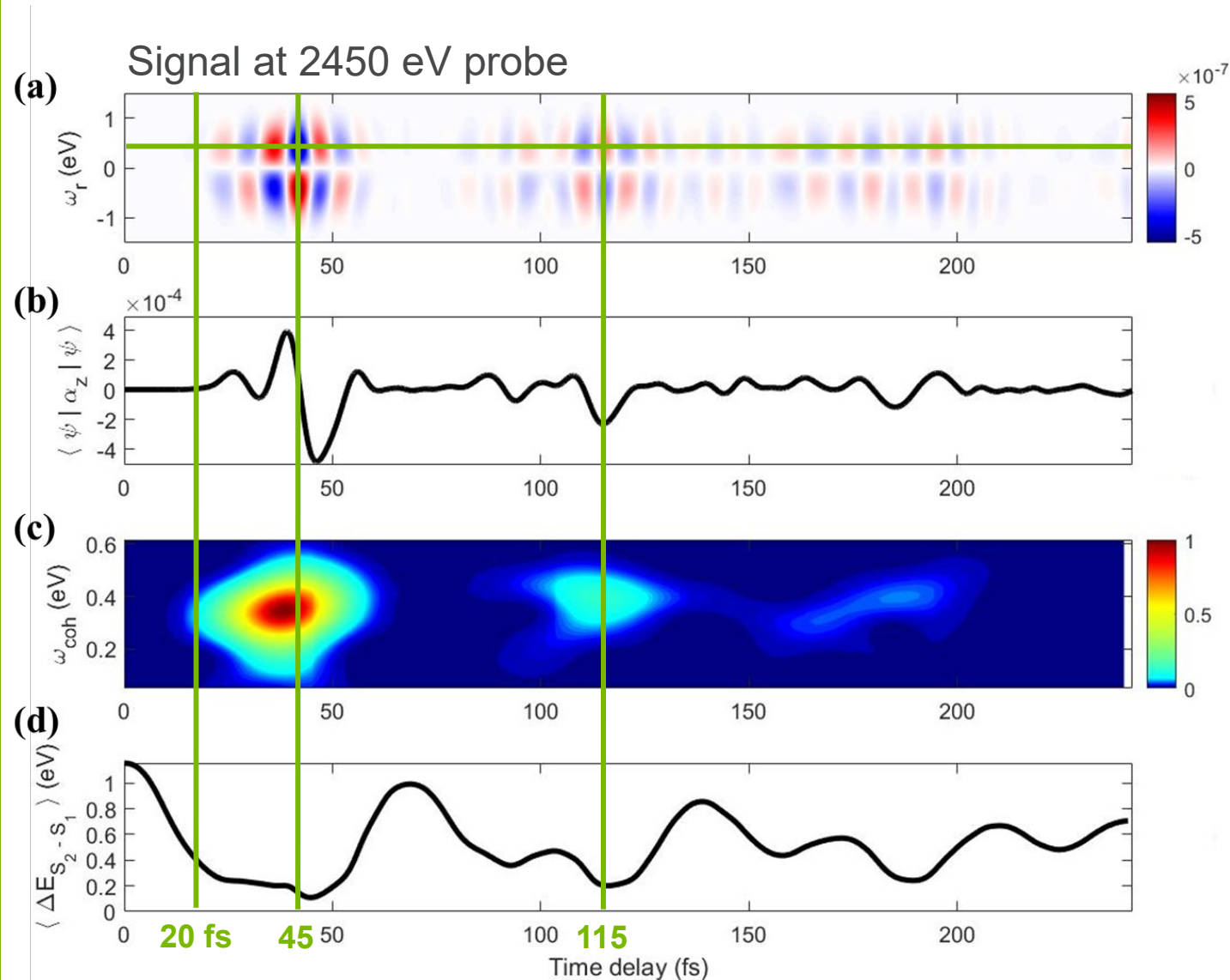
Population (real)

Coherence (real + imag)



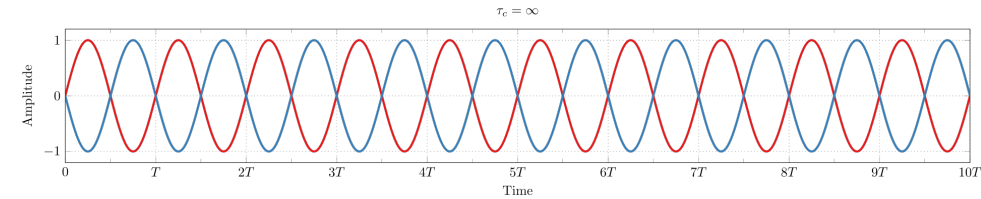
M. Kowalewski, K. Bennett, K. E. Dorfman, and S. Mukamel, *Phys. Rev. Lett.* **115**, 193003 (2015)

TRUECARS



$$\psi\psi^*$$

$$= c_1 c_1^* \phi_1 \phi_1^* + c_2 c_2^* \phi_2 \phi_2^* \quad \text{Population} \\ + c_1 c_2^* \phi_1 \phi_2^* + c_2 c_1^* \phi_2 \phi_1^* \quad \text{Coherence}$$



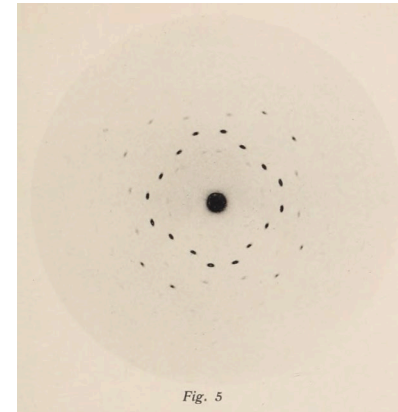
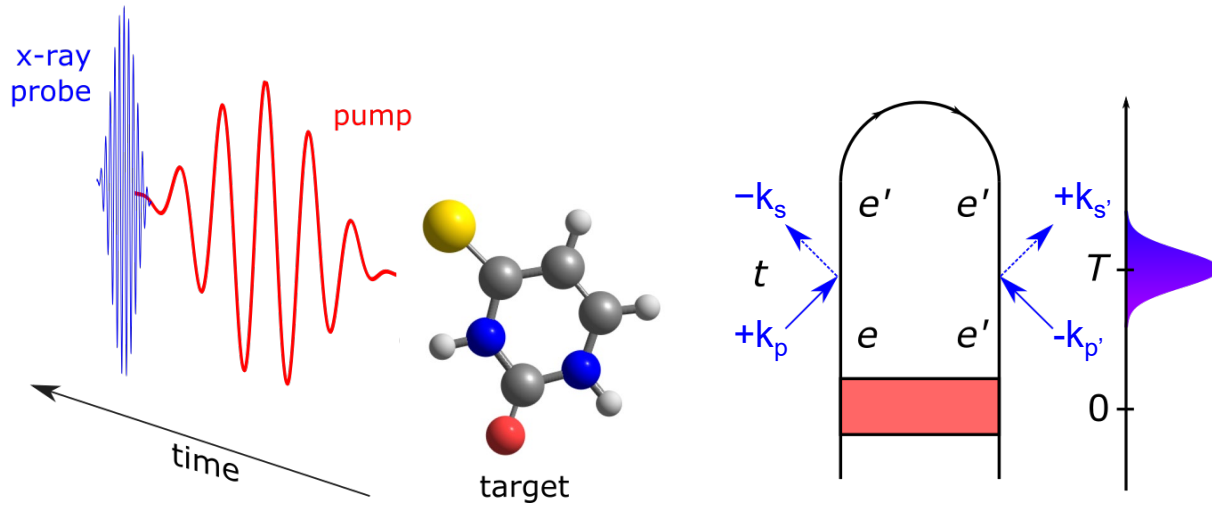
(a) TRUECARS signal. Coherence appears at 20 fs, strongest at 45 fs, oscillates with 70 fs period.

(b) Expectation value of transition polarizability between S_1 and S_2 .

(c) FROG spectrogram of the TRUECARS signal. Energy splitting is mapped.

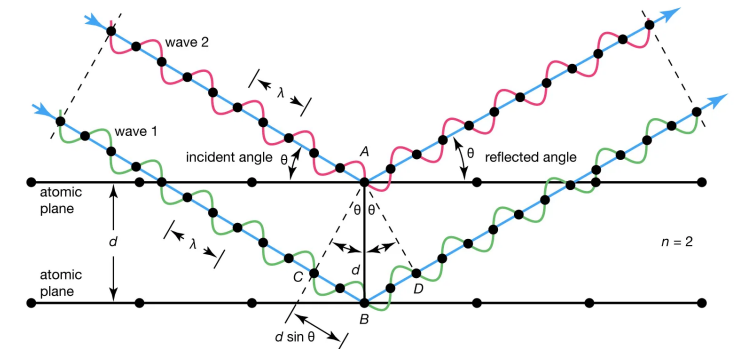
(d) S_2 - S_1 Energy splitting

Time-resolved X-ray diffraction (TRXD)



TRXD employs an **off-resonant x-ray probe pulses** which scatters off from electron densities between two electronic states e and e' .

- TRXD is based on scattering measurement, giving **spatial images**
- Traditionally, XD is used to probe crystal structure of sample at the ground state
- The signal can access **images of electron density evolving at the CI.**
- The mixed elastic/inelastic scattering event gives spectroscopic signatures from the non-adiabatic passages by positive/negative oscillations

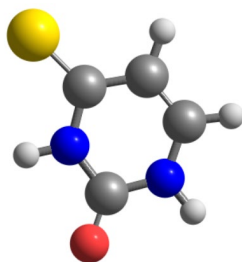


Bragg peak

$$S_2(\mathbf{q}, t) \propto F(\mathbf{q}) \int dt |E_p(t - T)|^2 |\rho_{gg} \langle \chi_g(t) | \hat{\sigma}_{gg} | \chi_g(t) \rangle|^2$$

$$F_{\mathbf{q}} = \sum_{\alpha} \sum_{\alpha \neq \beta} e^{-i\mathbf{q} \cdot (\mathbf{r}_{\alpha} - \mathbf{r}_{\beta})}$$

TRXD



$g=S_0$
 $e=S_1$ or S_2

Elastic scattering

- $\phi_1\phi_1 \rightarrow \phi_1\phi_1$
- $\phi_2\phi_2 \rightarrow \phi_2\phi_2$

$$S(\mathbf{q}, T) \propto N \int dt |E_p(t - T)|^2 \tilde{S}_1(\mathbf{q}, t),$$

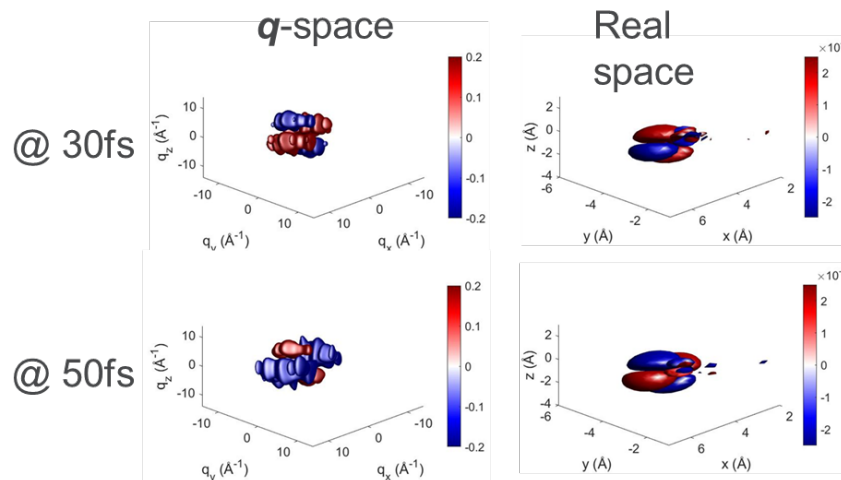
$$\begin{aligned} \tilde{S}_1(\mathbf{q}, t) = & \rho_{gg}(t) \langle \chi_g(t) | \hat{\sigma}_{gg}^\dagger \hat{\sigma}_{gg} | \chi_g(t) \rangle \\ & + \rho_{ee}(t) \langle \chi_e(t) | \hat{\sigma}_{ee}^\dagger \hat{\sigma}_{ee} | \chi_e(t) \rangle \\ & + \rho_{gg}(t) \langle \chi_g(t) | \hat{\sigma}_{ge}^\dagger \hat{\sigma}_{eg} | \chi_g(t) \rangle \\ & + \rho_{ee}(t) \langle \chi_e(t) | \hat{\sigma}_{eg}^\dagger \hat{\sigma}_{ge} | \chi_e(t) \rangle \\ & + 2\mathcal{R} \left[\rho_{eg}(t) \langle \chi_e(t) | \hat{\sigma}_{ee}^\dagger \hat{\sigma}_{eg} | \chi_g(t) \rangle \right. \\ & \left. + \rho_{eg}(t) \langle \chi_e(t) | \hat{\sigma}_{eg}^\dagger \hat{\sigma}_{gg} | \chi_g(t) \rangle \right], \end{aligned}$$

Inelastic scattering

- $\phi_1\phi_1 \rightarrow \phi_2\phi_2$
- $\phi_2\phi_2 \rightarrow \phi_1\phi_1$

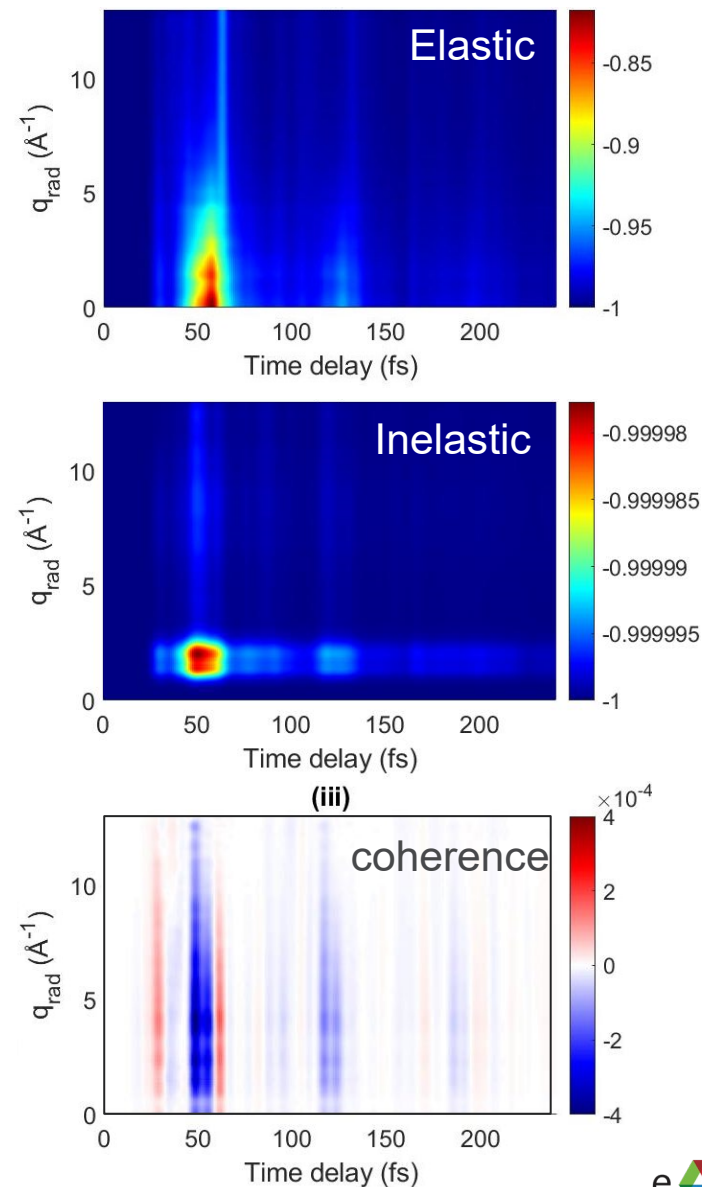
Coherence scattering!

- $\phi_1\phi_2 \rightarrow \phi_1\phi_1$
- $\phi_1\phi_2 \rightarrow \phi_2\phi_2$



Y. Nam et al., J. Phys. Chem. Lett. 2021, 12, 51, 12300

4-thiouracil signal



Summary

- Conical intersection dynamics in gas-phase 4-thiouracil was probed by 3 different X-ray techniques
- **ATAS** probes “**population**” of specific electronic state, $\pi\pi^*$ vs $n\pi^*$
- **TRUECARS** probes “**coherence**” created at the CI
- **TRXD** images the charge density evolving during the CI

Table 1. Comparison of the TRXAS, TRUECARS, and TRXD Techniques

technique	signal	pulse requirement	information	difficulties
TrXAS (established)	$S(\omega_s, T) = -\text{Im} \mu_{e'f} \mu_{fe}^* \frac{\epsilon_{pr}^*(\omega_s) \epsilon(\omega_s) \rho_{ee'}(T)}{\omega_s - \omega_{fe'} + i\gamma_{fe'}}$	single probe pulse	precise timing state-specific information vibrational coherence	population background overlapping absorption lines
TRUECARS (proposed)	$S(\omega_s, T) = \text{Im} \int dt \mathcal{E}_B^*(\omega_s) \mathcal{E}_N(t - T) e^{i\omega_s(t-T)} \langle \widehat{\alpha(t)} \rangle$	hybrid pulse (broadband/ narrowband)	energetic profiles vibronic coherence structure	requires two X-ray probe pulses (phase control)
TRXD	$S(\mathbf{q}, T) = N \int dt \mathcal{E}_X(t - T) ^2 \tilde{S}_1(\mathbf{q}, t)$	hard X-ray pulse	spatial information: electron transition densities at CoIns	extraction of coherences from the dominant population background

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Max D.J. Waters



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